NAAN MUDHALVAN | SMARTBRIDGE

Project Report

Auto Ticket Classification using Flow Designer

An automated IT helpdesk solution built on ServiceNow

Team ID : SWTID-2026-8965

Team Size: 5

Role	Name
Team Leader	Uvan M
Team Member	Uva Chandran N
Team Member	Vaahai Chezhiyan
Team Member	Varun S
Team Member	Veeresh Aarya V

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1. Introduction

Project Overview

The school IT helpdesk receives multiple incident requests every day from students and teachers. Typical requests relate to Wi-Fi and network connectivity, projector and hardware failures, forgotten passwords and slow computers. At present, IT staff must read each ticket manually and choose its category, which is time-consuming and inefficient.

Auto Ticket Classification using Flow Designer automates this step on the ServiceNow platform. When a ticket is created, a Flow Designer flow reads the keywords in the ticket's Short Description, assigns the correct Category and Subcategory, and sends an automatic confirmation email to the caller. The whole solution is built with Flow Designer and platform configuration, so it needs no custom scripting.

The project was carried out in structured phases: requirement analysis and planning, backend development and configuration, automation using Flow Designer with email notification, testing and validation, and finally deployment through an update set together with documentation.

Purpose

The purpose of the project is to remove manual effort from ticket triage and make routing of IT requests faster and more consistent. The specific objectives are:

- Automatically classify incidents at the time of creation.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Implement a no-code, easily maintainable solution.

2. Ideation Phase

Problem Statement

The school IT helpdesk receives multiple incident requests daily from students and teachers, such as Wi-Fi issues, projector failures, password problems and slow computers. IT staff currently review each request manually and assign a category, which is time-consuming and inefficient. For every ticket, staff must:

- Identify the issue type.
- Select the correct category and subcategory.
- Inform the caller that the ticket has been created.

This manual process is time-consuming, error-prone and not scalable as the number of requests grows.

Problem Statement	Description
I am	A student or teacher who needs IT support at school.
I am trying to	Report an issue (Wi-Fi, projector, password or slow computer) and get it to the right person quickly.
But	Every ticket has to be reviewed and categorised by hand by IT staff.
Because	The process is manual, time-consuming, error-prone and does not scale with the volume of requests.
Which makes me feel	Uncertain whether my request has been received and when it will be handled.

Empathy Map Canvas

The empathy map below summarises the two groups affected by the problem, using only the needs and pain points identified in the problem statement.

	Students / Teachers (Callers)	IT Staff (Agents)
Who	Raise IT tickets for Wi-Fi, projector, password and performance issues.	Review every incoming ticket and assign its category and subcategory.
Does	Enters the Caller and a Short Description of the issue.	Reads each request, selects category/subcategory, and informs the caller.
Pains	No quick acknowledgement that the request was received; categorisation depends on manual handling.	Time-consuming, error-prone manual review that is not scalable as requests increase.
Gains	Only the Caller and Short Description are needed; an automatic confirmation email is received.	Consistent, automatic classification and less manual effort; tickets arrive already categorised.

Brainstorming

Different ways of solving the classification problem were considered. The team selected a keyword-based, no-code approach because it is simple, transparent and easy to maintain in an educational setting.

Approach	Consideration	Decision
Manual classification (current process)	Time-consuming, error-prone, not scalable.	Rejected – this is the problem to be solved.
Script-based or machine-learning classification	More complex to build and maintain than the school environment needs.	Not used in this project; listed under Future Scope.
Keyword rules in Flow Designer	No-code, easy to read and modify; classifies at ticket creation.	Selected.

Key ideas selected: (1) a custom table to store ticket records in a structured format; (2) dependent Category/Subcategory choices so only valid combinations are possible; (3) a Flow Designer flow with If/Else If keyword conditions; (4) an automatic Send Email action; and (5) an update set so the work can be moved to another instance.

3. Requirement Analysis

Customer Journey Map

The journey of a student or teacher, from raising an issue to receiving confirmation, is shown below.

Stage	What the user does	What the system does	Outcome
1. Raise	Opens a new Incident Workflow record and enters the Caller and Short Description.	Generates a ticket number automatically (prefix INC).	Ticket is created.
2. Classify	Nothing – no need to choose a category.	Flow triggers on record creation where Category is empty and checks the Short Description for keywords.	Category and Subcategory are filled in automatically.
3. Acknowledge	Receives an email.	Send Email action emails the caller: “Your Request for the issue has been submitted”.	Caller knows the request was received.
4. Resolve	Waits for support.	IT agent works on a ticket that is already categorised.	Faster routing.

Solution Requirement

Business requirements

The system must:

1. Automatically classify IT tickets based on the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardised format.
6. Ensure easy maintenance and future scalability.

Data requirements – fields of the Incident Workflow table

Field Label	Type	Reference / Choices
Number	Auto Number	Prefix INC
Caller	Reference	sys_user
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Computer (dependent on Category)
Short Description	String	–
Description	String	–
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group
Assigned To	Reference	sys_user

Classification rules

If Short Description contains	Category	Subcategory
Wi-Fi or Network	Network	Wi-Fi
Projector	Hardware	Projector
Forgot password	Access	Forgot Password
Slow computer	Performance	Slow Computer

Data Flow Diagram

The data flow diagram shows how a ticket moves from the caller through the table, the flow and the email notification.

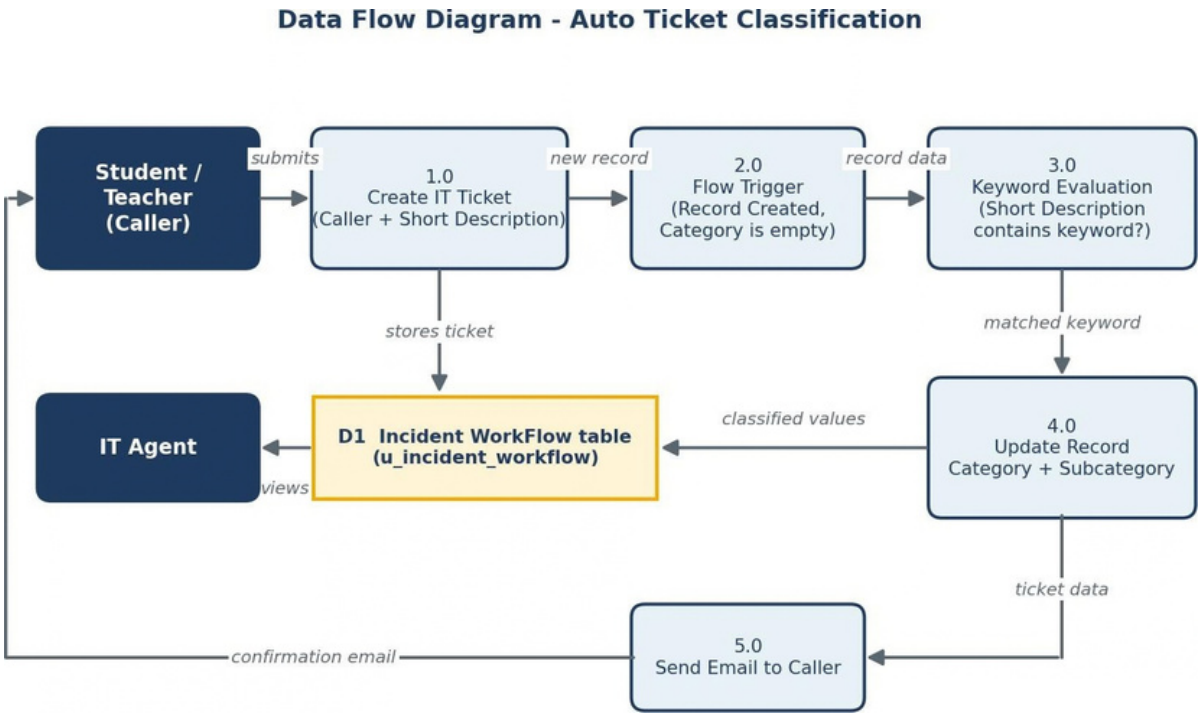


Figure 1: Data Flow Diagram – Auto Ticket Classification

Technology Stack

Component	Purpose in this project
ServiceNow platform (Global application scope)	Host platform. Work was done on a ServiceNow developer instance.
Tables and Form Designer	Create the custom table Incident Workflow (u_incident_workflow) and lay out its form.
Dictionary configuration	Dependent-field setting that links Subcategory choices to the selected Category.
Flow Designer (Workflow Studio)	No-code automation: trigger, If/Else If conditions, Update Record and Send Email actions.
Email (System Logs > Emails, SMTP)	Delivery and verification of the confirmation email.
Update Sets	Capture all configuration changes (Project Update Set) and export them as XML.

4. Project Design

Problem Solution Fit

Problem	Solution
Tickets are categorised by hand, which is slow and error-prone.	A Flow Designer flow reads the Short Description and sets Category and Subcategory automatically.
Invalid or inconsistent category combinations.	Subcategory is a dependent choice field, so only options belonging to the chosen Category are offered.
Callers are not told that their ticket exists.	A Send Email action notifies the caller as soon as the ticket is created.
Ticket data is unstructured.	A dedicated table with typed fields (reference, choice, string) and automatic numbering stores every ticket in a standard format.
Solution must be easy to maintain and extend.	No-code flow with clearly labelled conditions; work packaged in an update set.

Proposed Solution

A student or teacher enters only the Caller and the Short Description. On submission, the flow “Auto Classify School IT Tickets” starts. Its trigger is a record being created in the Incident Workflow table where Category is empty. The flow then evaluates the Short Description in sequence:

- If it contains **Wi-Fi** or **Network**: set Category = Network, Subcategory = Wi-Fi.
- Else If it contains **Projector**: set Category = Hardware, Subcategory = Projector.
- Else If it contains **Forgot password**: set Category = Access, Subcategory = Forgot Password.
- Else If it contains **Slow computer**: set Category = Performance, Subcategory = Slow Computer.

Finally, the Send Email action sends the caller a confirmation message with the subject “Your Request for the issue has been submitted”. This produces consistent classification, removes human error and reduces the effort required of IT staff.

Solution Architecture



Figure 2: Layered solution architecture

5. Project Planning & Scheduling

Project Planning

The project was executed in the phases below. Dates are shown only where they are recorded in the project screenshots.

Phase	Activities	Status / Evidence
Phase 1 – Requirement Analysis & Planning	Business use case and requirements; creation of the Project Update Set (Global) and setting it as the current set.	Completed. Update set created 27-09-2026.
Phase 2 – Backend Development & Configuration	Custom table Incident Workflow with auto-numbering; field creation and data types; Category–Subcategory dependency.	Completed. Table u_incident_workflow, form design and dependent field configured.
Phase 3 – Automation & Email Notification	Flow “Auto Classify School IT Tickets”: trigger, four classification branches, Send Email action.	Completed. Flow test run executed 28-09-2026.
Phase 4 – Testing, Validation & Security	Data validation checks; test scenarios; email verification in System Logs.	Completed. Email logged 28-09-2026.
Phase 5 – Deployment & Final Documentation	Project Update Set moved to Complete and exported to XML; project report prepared.	Completed. Update set state: Complete (38 customer updates).

Project team

Name	Role	Team ID
Uvan M	Team Leader	SWTID-2026-8965
Uva Chandran N	Team Member	SWTID-2026-8965
Vaahai Chezhiyan	Team Member	SWTID-2026-8965
Varun S	Team Member	SWTID-2026-8965
Veeresh Aarya V	Team Member	SWTID-2026-8965

6. Functional and Performance Testing

Performance Testing

Data validation checks

- Mandatory fields captured correctly.
- Auto-number generated without duplication.
- Category and Subcategory values stored accurately.
- Reference fields resolve correctly to the user and group tables.

Functional test cases

No.	Test	Expected result	Observed (evidence)
1	Create a new Incident Workflow record	A ticket number with prefix INC is generated automatically.	Record INC0001004 generated (Figure 7).
2	Select Category = Network	Subcategory list shows only the Wi-Fi option.	Dropdown offered None and Wi-Fi only (Figure 7).
3	Flow trigger on ticket creation	Flow runs when a record is created with an empty Category.	Test run shown as Completed (Figure 9).
4	Confirmation email	Caller receives “Your Request for the issue has been submitted”.	Send Email action completed; email logged on 28-09-2026 (Figures 9, 10, 11).
5	Test scenario: “WiFi not working in library”	Category = Network; Subcategory = Wi-Fi; email sent.	Email verified. Team documentation shows Network / Wi-Fi for a network issue (Figure 46); see note below for the recorded flow run.
6	Test scenario: “Projector not turning on”	Category = Hardware; Subcategory = Projector; email sent.	Team documentation shows Hardware / Projector (Figure 47); rule configured in the flow (Figure 8).

Flow execution performance

The execution details of the test run on 28-09-2026 (22:26:45) record the time taken by each step:

Step	Action	Type	Result	Time
1	If – SD is Wifi or Network	Flow Logic	Evaluated – False	96 ms
3	Else If – SD is Projector or Hardware	Flow Logic	Evaluated – False	12 ms
5	Else If – SD is Forgot password	Flow Logic	Evaluated – False	1 ms
7	Else If – SD is Performance	Flow Logic	Evaluated – False	0 ms
9	Send Email	Core Action	Completed	393 ms

7. Results

Output Screenshots

Backend configuration

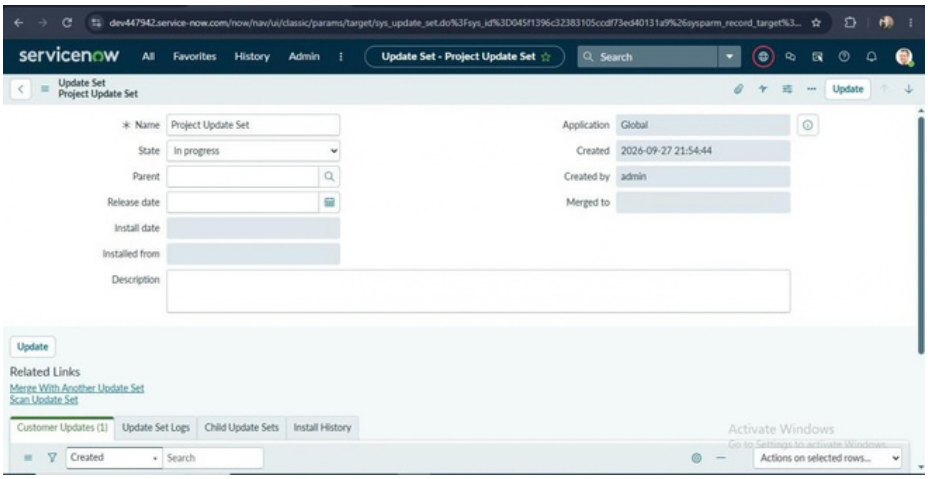


Figure 3: Project Update Set (Global) created and In progress – 27-09-2026

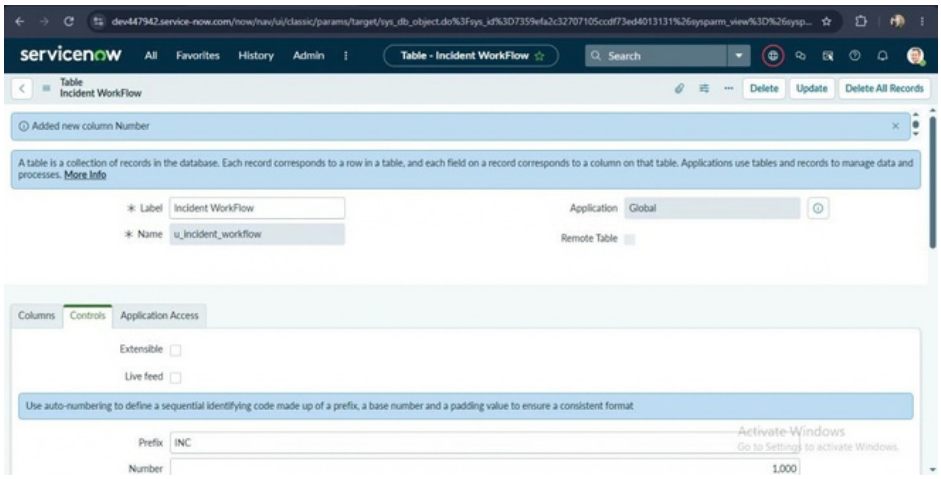


Figure 4: Custom table Incident Workflow (u_incident_workflow) with auto-numbering, prefix INC

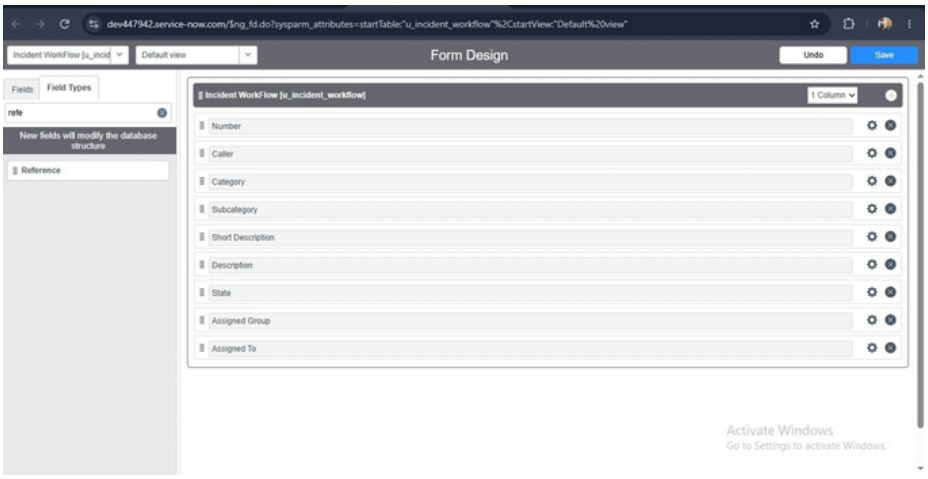


Figure 5: Form Design – Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group, Assigned To

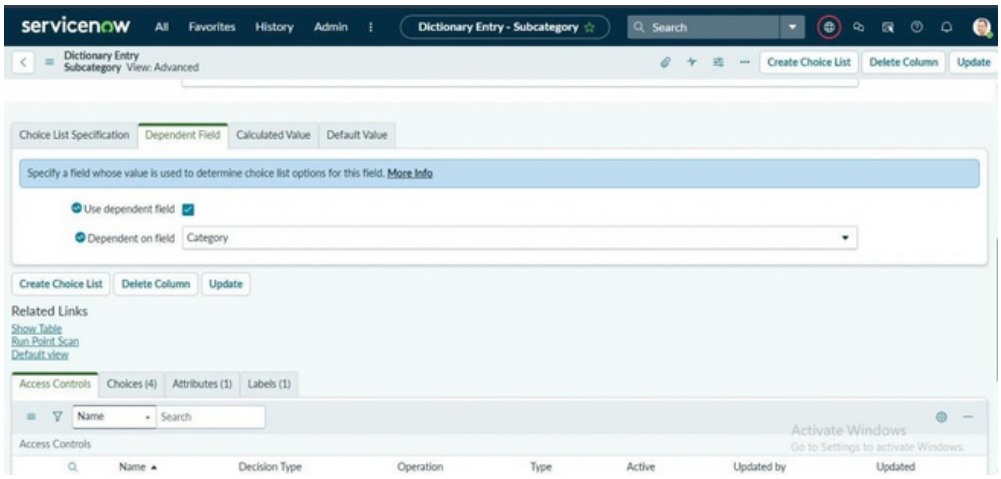


Figure 6: Subcategory dictionary entry – Use dependent field enabled, dependent on Category

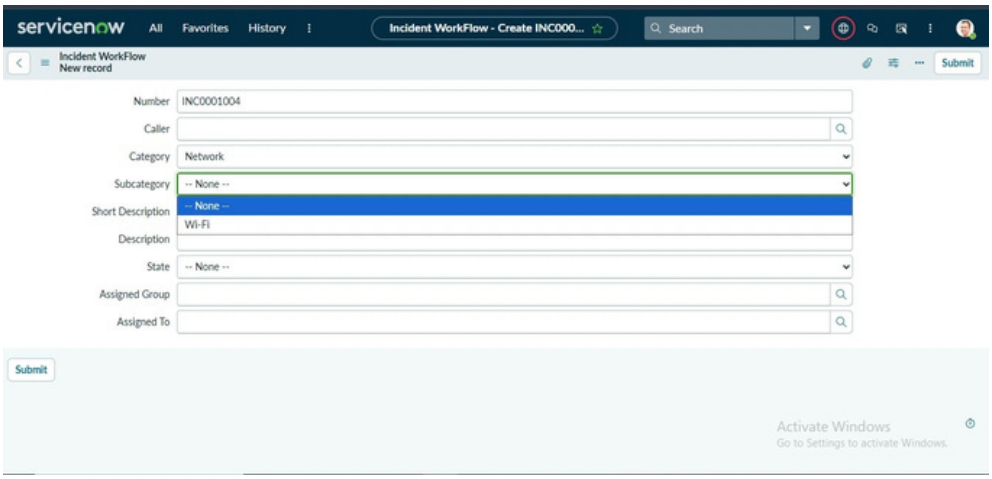


Figure 7: New record INC0001004 – Category Network offers the Wi-Fi subcategory

Automation using Flow Designer

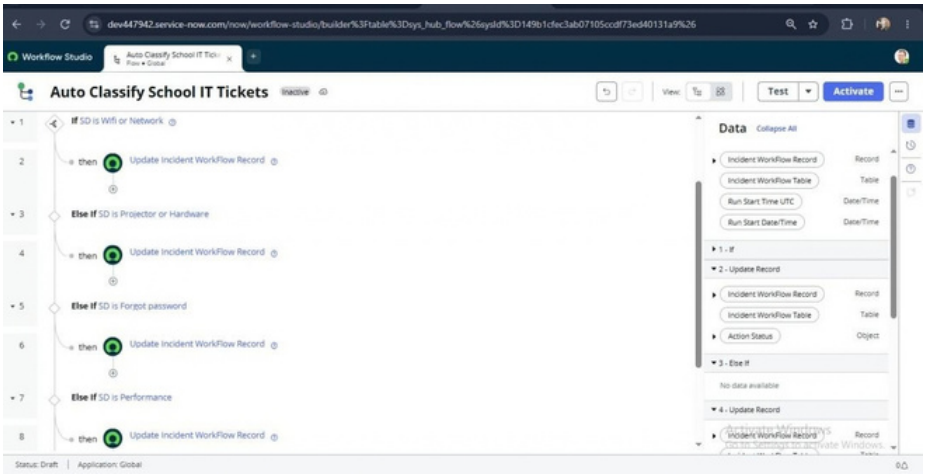


Figure 8: Flow "Auto Classify School IT Tickets" – If/Else If branches with Update Record actions (design view, status Draft)

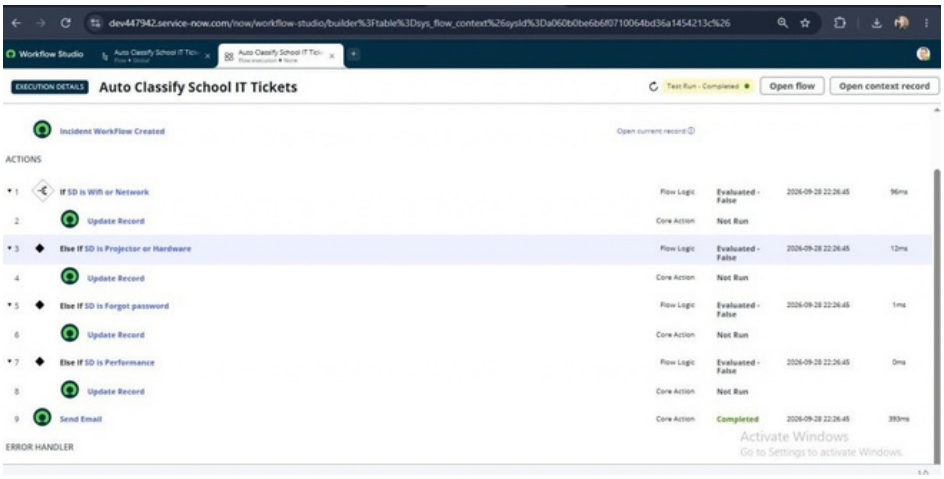


Figure 9: Flow execution details – Test Run Completed, Send Email completed in 393 ms

Email notification

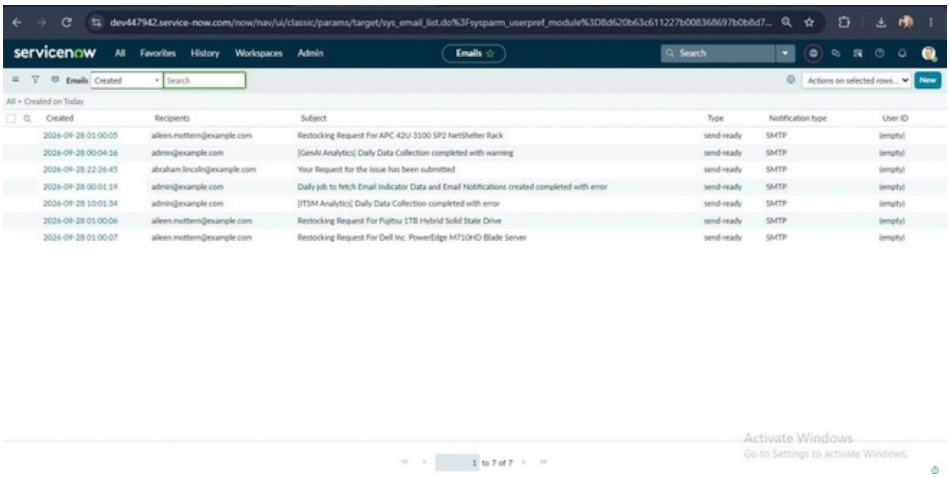


Figure 10: System Logs > Emails – "Your Request for the issue has been submitted" (28-09-2026)

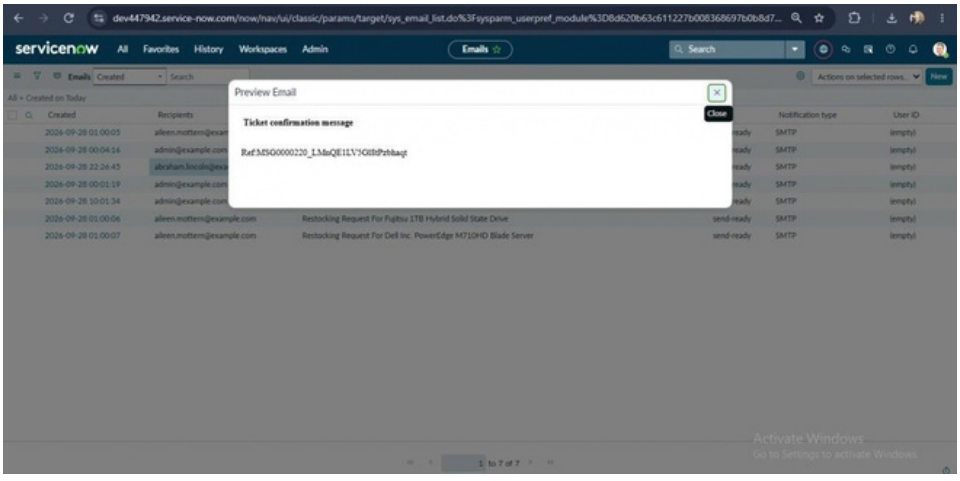


Figure 11: Email preview – "Ticket confirmation message"

Deployment

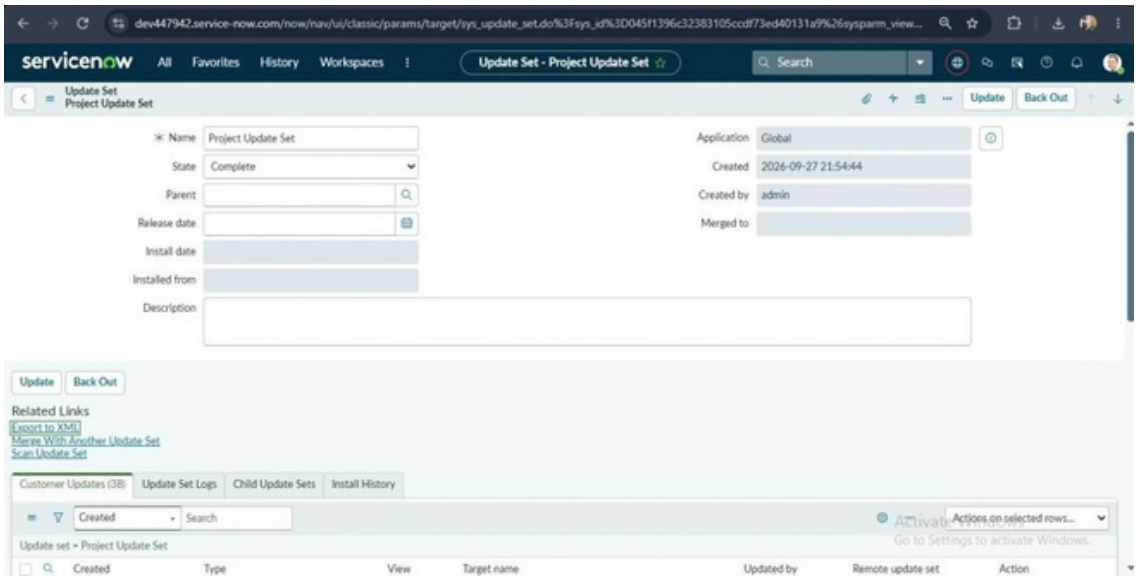


Figure 12: Project Update Set moved to Complete with 38 customer updates; Export to XML available

8. Advantages & Disadvantages

Advantages

- **Faster triage:** tickets are classified automatically at creation, with no manual review.
- **Consistency:** the same keyword always gives the same Category and Subcategory, removing human error.
- **Data integrity:** dependent choice fields prevent mismatched Category/Subcategory values.
- **Better communication:** callers get an immediate email acknowledgement.
- **No-code and maintainable:** built entirely in Flow Designer; rules can be read and changed easily.
- **Portable:** all changes are captured in an update set that can be exported as XML.

Disadvantages

- **Keyword dependent:** classification works only when the Short Description contains the configured text (for example “Projector” or “Slow computer”). Different wording will not match.
- **Limited categories:** only four issue types are handled; other issues need new branches.
- **No fallback branch:** a ticket that matches no keyword stays unclassified, although the confirmation email is still sent.
- **Basic scope:** assignment of tickets to groups or agents and SLA tracking are not part of this version.

9. Conclusion

The Auto Ticket Classification using Flow Designer project delivers an end-to-end automation for a school IT helpdesk. Through structured requirement analysis, careful backend configuration and no-code automation, tickets can now be classified from the Short Description supplied by the user, without manual categorisation by IT staff.

Dependent choice fields keep the data accurate, and the automated email improves communication by acknowledging every ticket on creation. The phased approach, with testing and an update set for deployment, kept the solution clear, stable and easy to maintain.

Overall, the project shows how ServiceNow can be used to build scalable, maintainable and user-friendly automation without complex scripting or machine learning. It is well suited to small and medium environments such as educational institutions.

10. Future Scope

The solution can be extended in several directions:

- **Assignment automation** – automatically set the Assigned Group and Assigned To based on the Category.
- **SLA tracking** – monitor response and resolution times for classified tickets.
- **Predictive intelligence** – move from fixed keywords to machine-learning-based classification.
- **Additional categories and keyword variants**, and a default branch for tickets that match no keyword.